





Objectives

- Understand what information is recorded in the Activity Logs.
- Understand how to control what information is written to the Activity Logs.
- Understand how to control the type of output data set used for the Activity Logs.
- Understand the diagnostic value of the Activity Logs.

Purpose of Activity Logs

- Contains information about the activity and errors that occur during HSM processing
- Used to determine if functions performed as expected
- Opened at HSM startup and closed at various points in HSM processing
- Contains data-set-level information controllable via SETSYS command
- Examined to determine the success or failure of a command or automatic function

The Activity Logs contain messages that record important events in HSM processing. Among the events captured are:

- Start of Automatic Processing
- Start of Volume Processing
- Data Sets Processed during Volume Processing
- End of Volume Processing
- End of Automatic Processing

Note: Each HSM image has its own set of activity logs

Notes:

Specifying Message Level for Recording Data Set Activity

SETSYS ACTLOGMSGGLVL(**FULL** | EXCEPTIONONLY
|REDUCED)

SETSYS ABARSACTLOGMSGGLVL(**FULL** | REDUCED)

DFSMSHsm 1.1.0 & 1.2.0

The SETSYS ACTLOGMSGVL controls which ARC0734I messages are written to the HSM Activity Logs. The ARC0734I message is issued for each data set during level- or volume-oriented processing or for each backup version processed as a result of the EXPIREBV command.

- FULL - ARC0734I written to the Activity Log during initial and subsequent movement of data sets, regardless of the success or failure of that movement. (Initial movement refers to data movement from an ML0 volume. Subsequent movement refers to data movement from an ML1 volume.
- EXCEPTIONONLY - ARC0734I written to the Activity Log during initial and subsequent movement of data sets ONLY when that movement fails.
- REDUCED - ARC0734I written to the Activity Log only during initial movement of data sets, regardless of the success or failure of that movement.

SETSYS ACTLOGMSGVL also affects which ARC0734I messages are written to the ARCLOGX data set.

The SETSYS ABARSACTLOGMSGVL is used to specify which DFSMSHsm messages are written to the respective ABARS activity log.

- FULL - specifies that all DFSMSdss messages are intercepted and written to the ABARS activity log.
- REDUCED - specifies that only DFSMSdss warning or error messages are displayed in the ABARS activity log. No informational messages will be written.

Notes:

Specifying the Type of Output Data Set Used for Activity Logs

SETSYS ACTLOGTYPE(**SYSOUT**(class) | DASD)

SETSYS ABARSACTLOGTYPE(**SYSOUT**(class) |
DASD)

DFSMSHsm 1.1.0 & 1.2.0

The SETSYS ACTLOGTYPE and ABARSACTLOGTYPE commands control whether the HSM or ABARS Activity Logs are written to SYSOUT or to dynamically-allocated DASD data sets.

- SYSOUT - Activity Logs written to spool data set. If no class is specified, the class is established by the SETSYS SYSOUT command, which defaults to A.
- DASD - Activity Logs dynamically allocated using a unit name of SYSALLDA using following naming convention:

HSMACT.Hhostid.funcname.Dyyddd.Thhmmss

where *funcname* = CMDLOG for the Command Activity Log
BAKLOG for the Backup Activity Log
DMPLOG for the Dump Activity Log
MIGLOG for the Migration Activity Log

- The names for ABARS Activity Logs are different in DFHSM and DFSMSHsm
 - ABACKUP
HSMACT.Hhostid.agname.Dyyddd.Thhmmss -- DFHSM
HSMACT.Hhostid.ABACKUP.agname.Dyyddd.Thhmmss -- DFSMSHsm
 - ARECOVER
HSMACT.Hhostid.ABRxxLOG.Dyyddd.Thhmmss -- DFHSM
HSMACT.Hhostid.ARECOVER.agname.Dyyddd.Thhmmss -- DFSMSHsm

Note: The xx in ABRxxLOG indicates the ABARS task id, xx can range for 1- 15.

Each RECYCLE command writes messages to SYSOUT and is not under the control of the SETSYS ACTLOGTYPE command. However, with the implementation of DFSMSHsm 1.2.0, RECYCLE now gives you the option of sending Activity Log messages to an ODS. Specifying SYSOUT directs it to the class specified in the SYSOUT statement.

Notes:

Recycle Activity Logs

1.2.0

RECYCLE OUTDATASET(dsn)

DFSMSHsm 1.2.0 introduced the RECYCLE ACTIVITY LOG. All output from the RECYCLE command will be directed to the RECYCLE ACTIVITY LOG. The recycle command log records all data movement activity that results from processing the RECYCLE command. Any error messages that the RECYCLE command processor issues are also written in this log

OUTDATASET(dsn) is a new optional parameter of the RECYCLE command specifying where to send the messages generated during RECYCLE processing. For (dsn), substitute the fully qualified name of the data set to receive the RECYCLE messages. .

If OUTDATASET(dsn) is not specified on the RECYCLE command, DFSMSHsm 1.2.0 sends all messages generated during RECYCLE processing to the SYSOUT specified by the SETSYS command.

Notes:

Viewing the Activity Logs

RELEASE HARDCOPY

RELEASE HARDCOPY specifies that HSM print any of the four activity logs that have data. Printing occurs when ACTLOGTYPE is specified as SYSOUT on the SETSYS command. If the activity logs are allocated to DASD, printing does not occur, updated activity logs are closed. After printing (or closing) the logs, HSM allocates new activity logs.

Notes:

Sample Command Activity Log

```
***** TOP OF DATA *****
***** DFHSM COMMAND LOG, TIME 16:13:52, DATE 94/10/27 *****
16:13:52 94/10/27 DFHSM COMMAND LOG, TIME 16:13:52, DATE 94/10/27
16:15:50 94/10/27 ARC0200I TRAP IN MODULE ARCRSTR, CODE=0004, DEBUG ALWAYS ADDED
16:16:37 94/10/27 ARC0200I TRAP IN MODULE ARCRSTR, CODE=0069, DEBUG ALWAYS ADDED
16:17:32 94/10/27 ARC0503I ALLOCATION ERROR, VOLUME = USRPAK, RETURN CODE=000C,
REASON CODE=039C, INFO CODE=0000
16:17:32 94/10/27 ARC0200I TRAP IN MODULE ARCALVOL, CODE=0012, DEBUG ALWAYS
ADDED
16:18:14 94/10/27 DFHSM COMMAND LOG, TIME 16:18:14, DATE 94/10/27
***** BOTTOM OF DATA *****
*****
```

Sample Backup Activity Log

```
DFHSM BACKUP LOG, TIME 14:37:37, DATE 94/09/23
ARC0720I AUTOMATIC BACKUP STARTING
ARC0570I AUTOMATIC BACKUP FOR ALL VOLUME(S) TERMINATED, RC=18, REASON=0
ARC0722I BACKUP STARTING ON VOLUME PRIM01(NONMS) AT 14:40:47 ON 94/09/16 SYSTEM,3090
ARC0728I VTOC FOR VOLUME PRIM01 COPIED TO DATA SET DFHSM.VTOC.T474014.VPRIM01.D94259 ON VOLUME MITG001
ARC0120I BACKUP VOLUME A00782 ADDED, RC=0000, REAS=0000
ARC0734I ACTION=BACK-UP FRVOL=PRIM01 TOVOL=A00782 TRACKS= 0 RC= 0, REASON= 0, AGE= 23, DSN=PAULD.ONE.TWO
PAGE 0001 5695-DF175 DFSMSdss V1r1.0 Data Set Services 94.259 14:41
ADR035I (SCH) -PRIME(06), INSTALLATION EXIT ALTERED BYPASS FAC CLASS CHK DEFAULT TO YES
ADR035I (SCH) -PRIME(03), INSTALLATION EXIT ALTERED WORKUNIT DEFAULT TO .
DUMP DATASET(INCLUDE(TEST.ONE.TEST1)) -
OUTDDNAME(SYS00051) CANCELERROR OPTIMIZE(1)
ADR101I RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'DUMP '
ADR109I RI01 (01), 94259 14:41.18 INITIAL SCAN OF USER CONTROL STATEMENTS COMPLETED.
ADR035I (001) -PRIME(50), INSTALLATION EXIT ALTERED TAPE BLOCK SIZE DEFAULT TO 32K-BYTES

ADR016I (001) -PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK
ADR006I (001) -SETUP(01), 94259 14:41:18 EXECUTION BEGINS
ADR801I (001) -DTDSC(01), DATA SET FILTERING IS COMPLETE. 1 OF 1 DATA
SETS WERE SELECTED: 0 FAILED SERIALIZATION AND 0 FAILED FOR
OTHER REASONS.
ADR454I (001) -DTDSC(01), THE FOLLOWING DATA SETS WERE SUCCESSFULLY PROCESSED
TEST.ONE.TEST1
ADR013I (001) -CLTSK(01), 94259 14:41:19 TASK COMPLETED WITH RETURN CODE 0000
ADR012I (SCH) -DSSU(01), 94259 14:41:19 DFSMSdss PROCESSING COMPLETE. HIGHEST RE TURN CODE IS 0000
ARC0734I ACTION=BACK-UP FRVOL=PRIM01 TOVOL=A00782 TRACKS= 0 RC= 0, REASON= 0, AGE= 2, DSN=TEST.ONE.TEST1
PAGE 0001 5695-DF175 DFSMSdss V1r1.0 Data Set Services 94.259 14:41
ADR035I (SCH) -PRIME(06), INSTALLATION EXIT ALTERED BYPASS FAC CLASS CHK DEFAULT TO YES
ADR035I (SCH) -PRIME(03), INSTALLATION EXIT ALTERED WORKUNIT DEFAULT TO .
DUMP DATASET(INCLUDE(TEST.ONE.TEST12)) -
OUTDDNAME(SYS00053) CANCELERROR OPTIMIZE(1)

ARC0734I ACTION=BACK-UP FRVOL=PRIM01 TOVOL=A00782 TRACKS= 1 RC= 0, REASON=0, AGE= 77,
DSN=HSMATH0.F11DCLAS.N02.PSFB
ARC0734I ACTION=BACK-UP FRVOL=PRIM01 TOVOL=A00782 TRACKS= 1 RC= 0, REASON=0, AGE= 77,
DSN=HSMATH0.F11DCLAS.N03.POU
ARC0734I ACTION=BACK-UP FRVOL=PRIM01 TOVOL=A00782 TRACKS= 0 RC= 0, REASON=0, AGE= 77,
DSN=HSMATH0.F11DCLAS.N25.GDG.MODEL
ARC0734I ACTION=BACK-UP FRVOL=PRIM01 TOVOL=A00782 TRACKS= 3 RC= 0, REASON=0, AGE= 77,
DSN=HSMATH0.F11DCLAS.N04.DAF
ARC0734I ACTION=BACK-UP FRVOL=PRIM01 TOVOL=A00782 TRACKS= 0 RC= 0, REASON=0, AGE= 77,
DSN=HSMATH0.F11DCLAS.N45.GDG.MODEL
ARC0734I ACTION=BACK-UP FRVOL=PRIM01 TOVOL=A00782 TRACKS= 0 RC= 0, REASON=0, AGE= 77,
DSN=HSMATH0.F11DCLAS.N65.GDG.MODEL
ARC0723I BACKUP ENDING ON VOLUME PRIM01 AT 14:41:29, 13 DATA SETS BACKED UP
ARC0721I AUTOMATIC BACKUP ENDING
```

Sample Migration Activity Log

```
ARC0520I PRIMARY SPACE MANAGEMENT STARTING
ARC0570I PRIMARY SPACE MANAGEMENT FOR ALL VOLUME(S) TERMINATED, RC=18, REASON=0

ARC0522I SPACE MANAGEMENT STARTING ON VOLUME PRIM01(NONMS) AT 08:53:13 ON 94/09/22, SYSTEM 3090
ARC0734I ACTION=MIGRATE FRVOL=PRIM01 TOVOL= *** TRACKS= 0 RC= 30, REASON=22, AGE= 29, DSN=PAULD.ONE.TWO
PAGE 0001 5695-DF175 DFSMSdss V1R1.0 Data Set Services 94.265 08:53
ADR035I (SCH)-PRIME(06), INSTALLATION EXIT ALTERED BYPASS FAC CLASS CHK DEFAULT TO YES
ADR035I (SCH)-PRIME(03), INSTALLATION EXIT ALTERED WORKUNIT DEFAULT TO .
DUMP DATASET(INCLUDE(TEST.ONE.TEST1)) -
OUTDDNAME(SYS00065) CANCELERROR OPTIMIZE(1)
ADR101I RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'DUMP '
ADR109I RI01 (01), 94265 08:53:15 INITIAL SCAN OF USER CONTROL STATEMENTS
ADR035I (001)-PRIME(50), INSTALLATION EXIT ALTERED TAPE BLOCK SIZE DEFAULT TO 32, COMPLETED.
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK K-BYTES
ADR006I (001)-SETUP(01), 94265 08:53:15 EXECUTION BEGINS

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SETS WERE SELECTED: 0 FAILED SERIALIZATION AND 0 FAILED FOR
OTHER REASONS.
ADR454I (001)-DTDSC(01), THE FOLLOWING DATA SETS WERE SUCCESSFULLY PROCESSED
TEST.ONE.TEST1
ADR013I (001)-CLTSK(01), 94265 08:53:15 TASK COMPLETED WITH RETURN CODE 0000
ADR012I (SCH)-DSSU(01), 94265 08:53:15 DFSMSdss PROCESSING COMPLETE. HIGHEST RE
ARC0734I ACTION=MIGRATE FRVOL=PRIM01 TOVOL=MIG001 TRACKS= 1 RC= 0, REASON=TURN CODE IS 0000
PAGE 0001 5695-DF175 DFSMSdss V1R1.0 Data Set Services 94.265 0, AGE= 8, DSN=TEST.ONE.TEST1
ADR035I (SCH)-PRIME(06), INSTALLATION EXIT ALTERED BYPASS FAC CLASS CHK DEFAULT 08:53
ADR035I (SCH)-PRIME(03), INSTALLATION EXIT ALTERED WORKUNIT DEFAULT TO . TO YES
DUMP DATASET(INCLUDE(TEST.ONE.TEST10)) -
OUTDDNAME(SYS00069) CANCELERROR OPTIMIZE(1)
ADR101I RI01 (01), TASKID 001 HAS BEEN ASSIGNED TO COMMAND 'DUMP '
ADR109I RI01 (01), 94265 08:53:17 INITIAL SCAN OF USER CONTROL STATEMENTS
ADR035I (001)-PRIME(50), INSTALLATION EXIT ALTERED TAPE BLOCK SIZE DEFAULT TO 32, COMPLETED.
ADR016I (001)-PRIME(01), RACF LOGGING OPTION IN EFFECT FOR THIS TASK K-BYTES

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OTHER REASONS.
ADR454I (001)-DTDSC(01), THE FOLLOWING DATA SETS WERE SUCCESSFULLY PROCESSED
TEST.ONE.TEST4
ADR013I (001)-CLTSK(01), 94265 08:53:22 TASK COMPLETED WITH RETURN CODE 0000
ADR012I (SCH)-DSSU(01), 94265 08:53:22 DFSMSdss PROCESSING COMPLETE. HIGHEST RETURN CODE IS 0000
ARC0734I ACTION=MIGRATE FRVOL=PRIM01 TOVOL=MIG001 TRACKS= 1 RC= 0, REASON= 0, AGE= 8, DSN=TEST.ONE.TEST4
ARC0400I VOLUME PRIM01 IS 72% FREE, 000018 FREE TRACK(S), 00006 FREE CYLINDER(S) 0, AGE= 8, DSN=TEST.ONE.TEST4
ARC0401I LARGEST EXTENTS FOR PRIM01 ARE CYLINDERS 6, TRACKS 90 2 1
ARC0402I VTOC FOR PRIM01 IS 0002 TRACKS(00106 DSCBS), 00091 FREE DSCBS(85% OF TOTAL)
ARC0523I SPACE MANAGEMENT ENDED ON VOLUME PRIM01, 5 DATA SET(S) MIGRATED/DELETED, 5 TRACK(S) FREED, MINAGE 8,
TIME 08:53:23
ARC0400I VOLUME MIG001 IS 87% FREE, 000041 FREE TRACK(S), 00006 FREE CYLINDER(S) 103 26 FRAG .148
ARC0401I LARGEST EXTENTS FOR MIG001 ARE CYLINDERS 6, TRACKS 103 26 FRAG .148
ARC0402I VTOC FOR MIG001 IS 0002 TRACKS(00106 DSCBS), 00089 FREE DSCBS(83% OF TOTAL)
ARC0521I PRIMARY SPACE MANAGEMENT ENDED SUCCESSFULLY
DFHSM MIGRATION LOG, TIME 08:53:24, DATE 94/09/22
```

Where in the World Did HSM Put those Messages?



For the most part, routing of messages to the appropriate Activity Log is intuitive. Messages related to Migration are written to the Migration Activity Log, messages resulting from Dump activity are written to the Dump Activity Log, etc. Finding certain messages, however, is not quite as obvious.

Messages written to the Command Activity Log regardless of the function:

- ARC0503I ALLOCATION ERROR - Produced when HSM has attempted to allocate a data set or volume dynamically and the allocation failed.
- ARC0200I or ARC0208I TRAP IN MODULE / TRAP FOR ERRORCODE Issued by a user set trap or by the invocation of the error processing routine to perform a specific action when a certain errcode is produced.

Messages relating to CDS Backup are written to either the Command or Backup Activity Log, depending on your maintenance level. With the release of DFSMSHsm 1.2.0, all messages relating to CDS backup are placed in the Backup and Command Activity Log.

Notes:

Diagnostic Value of Activity Logs

The HSM Activity Logs contain messages useful for describing problems you may be experiencing. These messages contain return and reason codes which can be used when performing searches in IBMLink. In addition, the messages provide the context for failures you may have experienced. For example, if you find that data sets are not being processed, check to see if the volume they reside on was skipped.